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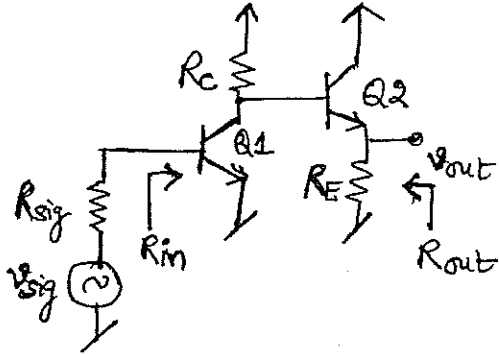
Degree: B.Tech. Semester: 3rd — Course work
MID-SEMESTER EXAMINATION, September, 2024

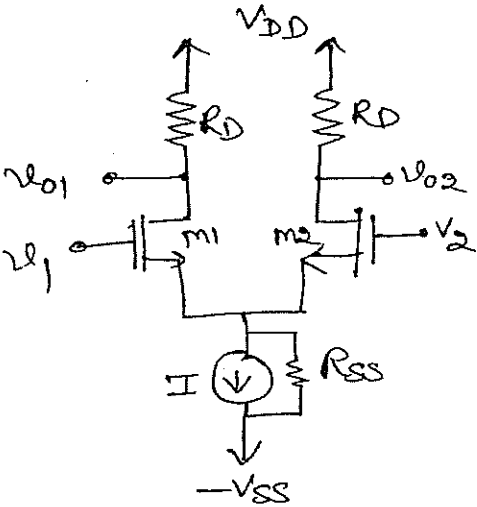
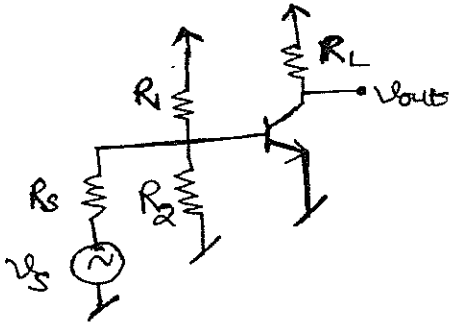
Course Title: **Microelectronics**
 Course Code: **ECECC07**

Duration: 1:30 Hours

Max. Marks: 15

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	For the multistage amplifier shown in Fig. 1, draw its small signal equivalent circuit and Find the expressions for A_v, R_{in} and R_{out}. Consider $V_A = \infty$.  Fig. 1	2.5	CO1
1b	Draw the circuit of BJT based Darlington pair and discuss its advantages.	0.5	CO1
2a	Draw the circuit diagram of a BJT-based differential amplifier and derive the expression for the large signal collector currents (I_{C1} & I_{C2}) of both BJTs. Additionally, plot the graph of differential voltage versus collector currents for the same.	2.5	CO1
2b	Define CMRR and how it can be improved in case of single ended output?	0.5	CO1
3a	The MOS amplifier shown in Fig. 2 has $g_m = 4\text{mS}$ and $V_A = \infty$. If $ A_{DM} = 100$ and $\text{CMRR} = 80\text{dB}$, find R_{SS} and R_D .	2.5	CO1

	 Fig. 2		
3b	State and prove Miller's theorem.	0.5	CO1
4a	Derive the expressions for approximate high cut-off frequency for the CE amplifier shown in Fig. 3.  Fig. 3	2.5	CO2
4b	Draw the 4-terminal high frequency model of MOSFET showing all parasitic capacitances.	0.5	CO2
5a	Draw the Bode plots for magnitude and phase response for the transfer function given below: $H(s) = \frac{50(s+40)}{(s+20)(s+60)}$	2.5	CO2
5b	Draw the BJT based differential amplifier circuit of differential to single ended converter using active load and give the expression of \$A_D\$.	0.5	CO1

-----GOOD LUCK-----